Forecasting Germany's Real GDP Growth

ESS Econometrics (20203) – Group Assignment, Part II

Quarterly sample: 2000Q1–2025Q4 | Forecast evaluation: 2016Q1–2025Q4

Bocconi University | May 2026

Question

Which dynamic specification best explains and forecasts Germany's year-on-year real GDP growth?

Answer in one line

The selected ADL is the best individual econometric model; an equal-weight forecast combination gives the lowest overall forecast error.

Models compared: BIC-selected ADL, AR(2), VAR(2), CCI leading-indicator regression, and an equal-weight forecast combination.

Empirical set-up

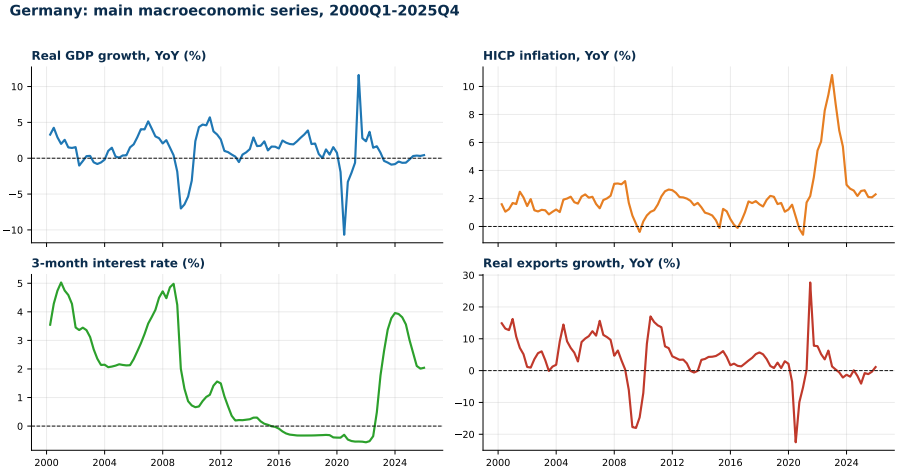
Data and variables

- Country: **Germany**.
- Quarterly sample: **2000Q1–2025Q4** (104 observations before lagging).
- Target: g_t, real GDP growth, year-on-year %.
- Predictors: HICP inflation π_t, short-term interest rate i_t, real exports growth, and consumer confidence (CCI).
- Transformations: GDP, inflation, and exports are in YoY rates; short rate and CCI are kept in levels.

Forecasting design

- Evaluation window: **2016Q1–2025Q4**.
- Horizon: **1-quarter-ahead** point forecasts.
- Estimation window: recursive / expanding.
- Software: Python; OLS and VAR estimation.

Descriptive macro picture



- GDP and exports display the largest crisis swings, especially around 2008–2009 and 2020–2021.
- Inflation rises sharply after 2021; interest rates tighten later, after a long near-zero period.

Descriptive statistics: scale and co-movement

Variable	Mean	Std. dev.	Min	Max
GDP growth g_t	1.11	2.66	-10.68	11.62
Inflation π_t	2.07	1.88	-0.59	10.82
Short rate i_t	1.63	1.78	-0.57	5.03
Exports growth x_t	3.77	7.24	-22.52	27.69
CCI	100.00	1.33	96.75	101.98

Notes: 2000Q1–2025Q4. GDP, inflation, exports and the short rate are in percentage points; CCI is an index.

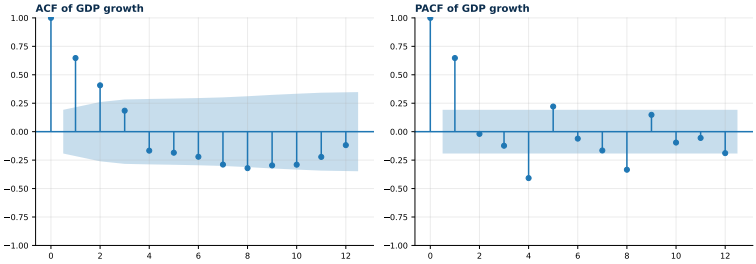
Correlation with GDP growth

Variable	Corr. with g_t
Exports growth	**0.90**
CCI	0.47
Inflation	0.10
Short rate	0.00

Implication

Exports and GDP move together contemporaneously, but forecasting requires lagged information. Inflation and rates look weak contemporaneously, so their value is tested through lags rather than one-period correlations.

Persistence in GDP growth



Model implication

- GDP growth has clear short-run persistence.
- PACF highlights a dominant first lag; the ACF decays gradually.
- This motivates AR and ADL dynamics rather than a static regression.
- Stationarity checks support modeling GDP in growth rates: ADF $p < 0.001$, KPSS $p = 0.10$.

Model specification strategy

Model family	Specification used	Role in the analysis
AR(2)	$g_t = \alpha + \phi_1 g_{t-1} + \phi_2 g_{t-2} + u_t$	Required benchmark based only on GDP persistence.
ADL grid	GDP lags $\{1, 2\}$; exports, inflation and rate lags $\{0, 1, 2\}$; BIC selects the final lag structure.	Main interpretable forecasting model.
VAR(2)	$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t$, with $y_t = (g_t, \pi_t, i_t)'$.	Required multivariate benchmark with GDP, inflation and the short rate.
CCI rule	One-quarter-ahead leading-indicator forecast based on consumer confidence.	Soft-information benchmark.
Combination	Equal-weight average of AR(2), ADL, VAR(2), and CCI forecasts.	Simple insurance against picking the wrong single model.

Selection principle

Models are selected in-sample using parsimony and BIC, then judged out-of-sample using recursive one-step-ahead forecast accuracy.

ADL lag selection by BIC

Rank	p_g	p_x	p_π	p_i	BIC
1	**1**	**0**	**2**	**2**	**201.67**
2	2	0	2	0	201.91
3	2	0	2	2	202.94
4	1	0	2	0	203.42
5	1	2	2	0	204.21

p_g, p_x, p_π, p_i denote the maximum included lags of GDP growth, export growth, inflation, and the short rate. Ranking uses the initial pre-forecast training sample.

Selected ADL

$$g_t = \alpha + \phi_1 g_{t-1} + \beta_1 \pi_{t-1} + \beta_2 \pi_{t-2} + \gamma_1 i_{t-1} + \gamma_2 i_{t-2} + u_t.$$

Why exports drop out

Exports are highly correlated with GDP in the same quarter, but lagged exports do not improve BIC once GDP persistence, inflation lags and rate lags are included.

Preferred ADL: estimates and hypotheses

Regressor	Coef.	p-value	Interpretation
Constant	1.337	< 0.001	baseline growth
g_{t-1}	**0.478**	< 0.001	GDP persistence
π_{t-1}	0.634	0.024	short-run inflation effect
π_{t-2}	-0.985	0.001	reversal after two quarters
i_{t-1}	1.881	0.003	rate block matters
i_{t-2}	-1.915	0.002	delayed reversal

Fit on the post-lag full sample: $R^2 = 0.534$, adjusted $R^2 = 0.510$, BIC $= 437.6$, $n = 102$.

Economic hypotheses

Null hypothesis	Test	p-value
No GDP persistence	$\phi_1 = 0$	0.001
No inflation content	$\beta_1 = \beta_2 = 0$	< 0.001
No rate content	$\gamma_1 = \gamma_2 = 0$	0.001

HAC Wald tests with four lags reject all three restrictions at the 1% level.

- The model is interpretable: persistence plus macro-financial lags.
- The lag signs should be interpreted as predictive dynamics, not as structural causal effects.

Diagnostics for the preferred ADL

Diagnostic	Statistic	p-value	Interpretation
Breusch–Godfrey LM(4)	17.87	**0.001**	serial correlation
White test	44.21	**0.001**	heteroskedasticity
Jarque–Bera	1202.84	< 0.001	non-normal residuals
Durbin–Watson	1.94	–	no obvious AR(1) issue

What this means

The ADL is useful for forecasting, but the residuals are not textbook-clean. Crisis quarters create fat tails and instability, which limits the reliability of classical OLS assumptions.

How we handle it

- Report economic hypotheses using HAC covariance.
- Avoid causal over-interpretation of individual coefficients.
- Rank models primarily through out-of-sample forecasts over 2016Q1–2025Q4.

Benchmark check

AR(2) has a similar Durbin–Watson statistic, but it still performs worse by RMSFE and MSFE in the forecast evaluation.

Recursive 1-step-ahead forecasting design



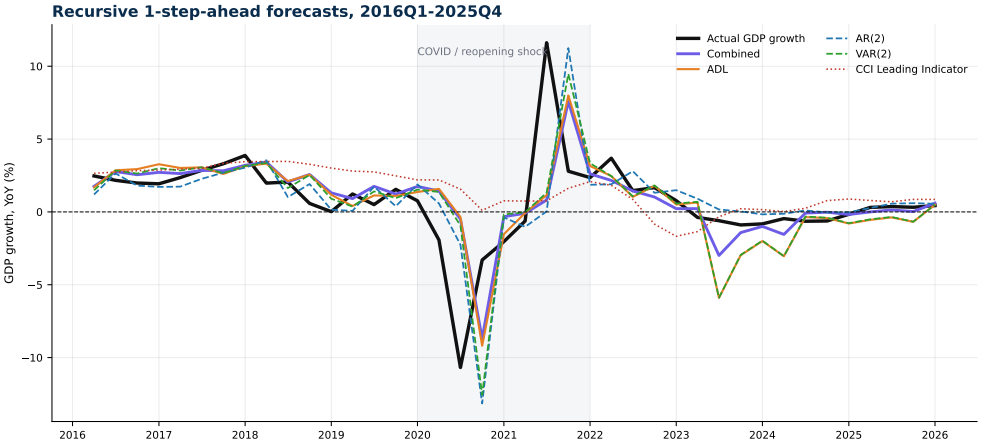
Evaluation window

- Last 10 years of the sample: **40 quarterly forecasts**.
- Forecasts evaluated against realized g_t from 2016Q1 to 2025Q4.

Accuracy metrics

- MSFE and RMSFE penalize large misses heavily.
- MAFE measures average absolute error.
- Bias is the mean forecast error: actual minus forecast.

Recursive forecasts: actual vs models



- All models struggle with the COVID trough and rebound; the combined forecast dampens some model-specific overreaction.
- The CCI-only forecast is smoother but misses sharp turning points.

Forecast accuracy: ranking and benchmark comparison

Model	MSFE	RMSFE	MAFE	Bias
Combined	**7.79**	**2.79**	**1.46**	-0.14
ADL	8.78	2.96	1.71	0.10
CCI leading indicator	9.13	3.02	1.78	-0.72
VAR(2)	10.16	3.19	1.85	0.15
AR(2)	10.20	3.19	1.68	-0.07

Forecast errors are in percentage points of year-on-year real GDP growth.

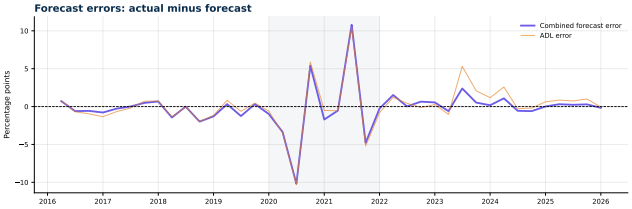
Main ranking

- **Combined forecast** is best on MSFE, RMSFE and MAFE.
- **ADL** is the best individual econometric model by MSFE/RMSFE.
- ADL lowers RMSFE by about **7.2%** relative to AR(2) and **7.0%** relative to VAR(2).

Why AR(2) still shows up

AR(2) is competitive by MAFE, but its larger misses are punished by RMSFE. This is why we report more than one error metric rather than relying on a single criterion.

Forecast error diagnostics



Unbiasedness and calibration

Model	Mean-error p	MZ joint p
Combined	0.761	< 0.001
ADL	0.837	< 0.001
VAR(2)	0.777	< 0.001
AR(2)	0.891	< 0.001
CCI indicator	0.133	0.012

Diebold–Mariano tests

Against the combined forecast, equal predictive accuracy is **not rejected** at 5%:

- Combined vs ADL: $p = 0.125$.
- Combined vs VAR(2): $p = 0.157$.
- Combined vs AR(2): $p = 0.300$.

Mean-error test: H_0 of zero average forecast error. MZ test: $g_t = \alpha + \beta \hat{g}_t + e_t$, $H_0 : \alpha = 0, \beta = 1$.

Economic interpretation and conclusion

Main conclusions

- Germany's GDP growth is persistent: the first lag is economically and statistically important.
- Inflation and short-rate lags jointly improve the selected ADL, consistent with delayed macro-financial transmission.
- Exports are highly correlated with GDP contemporaneously, but lagged exports are not selected by BIC once dynamics and macro-financial variables are included.
- The ADL outperforms the required AR(2) and VAR(2) benchmarks by MSFE/RMSFE; the equal-weight forecast combination performs best overall.

Caveat

The ranking is numerically clear but not statistically decisive under Diebold–Mariano tests. Diagnostics also warn against over-interpreting OLS assumptions, so the strongest claim is forecasting performance, not structural causality.

Transparency on LLM use

How LLM tools helped

- Structured the empirical workflow and presentation storyline.
- Checked that variable definitions, lags, and forecast metrics were described consistently.
- Helped improve code readability and slide wording.
- Suggested clearer ways to explain diagnostics and benchmark comparisons.

What remained our work

- We collected and transformed the data.
- We estimated the AR, ADL, VAR, CCI, and combined forecasts.
- We computed all tables, figures, tests, and accuracy measures.
- We verified the final numbers against the Python outputs.

Bottom line: LLMs supported organization and communication; the empirical results and final interpretation were produced and checked by the group.